CS F364: Design & Analysis of Algorithms

Quiz 1 – June 2, 2026

Coverage: Lectures 1–8

> Note
>
> **Instructions:**
>
> - Time: 15 minutes.
> - Write your answers clearly on paper, scan, and upload to Google Classroom.
> - Show all steps for full credit.

Question

Consider a divide-and-conquer algorithm `MyAlg` that operates on an array of size n.

- **Divide:** Split the array into **three** equal parts, each of size $n/3$.
- **Conquer:** Recursively process any **two** of the three parts. The third part is discarded.
- **Combine:** Merge the results of the two recursive calls by scanning through the *entire* original array once, taking $\Theta(n)$ time.

Assume n is a power of 3 and the base case is $T(1) = \Theta(1)$.

(a) [**1 Mark**] Write the recurrence relation for $T(n)$.

(b) [**3 Marks**] Solve the recurrence using the **recursion tree method**.

- Draw the recursion tree (describe its structure).
- Compute the cost at each level.
- Determine the number of levels.
- Sum the costs to obtain the total running time.

(c) [**1 Mark**] State the final asymptotic bound in Θ notation.

Bonus [**2 Marks**]**:** If `MyAlg` instead recursively processed **all three** parts (while keeping the same $\Theta(n)$ combine step), what would the new recurrence be, and what would its solution be?